

needs to be amplified before it is fed to the speakers.

Audio amplifiers are also classified as analogue and digital. Amplifiers that take analogue inputs and generate analogue outputs are called analogue audio amplifiers. These are usually referred to as Class AB. Digital audio amplifiers are Class D audio power amplifiers.

However, Class D amplifiers are not digital in true sense. These are switching amplifiers that work on pulse-width modulation (PWM) technique. The output of the Class D amplifier is in the form of pulses whose width is controlled by the amplitude and frequency of the signals. A low-pass filter is normally used to change the pulsating PWM signal into a smooth analogue signal before it is fed to the speaker.

Size does matter

The form factor of an electronic product is an important design consideration. So for the audio sub-systems, the designers need to select components that help to provide high quality and efficiency yet keep the size of the product small.

Class D amplifiers are 80-90 per cent efficient compared to Class AB amplifiers that are 50 per cent efficient. As Class AB amplifiers dissipate more heat, these require a bigger heat-sink. Using a high-efficiency Class D amplifier almost eliminates the need for a heat-sink. This helps to reduce the size of the audio system, making Class D amplifiers suitable for compact electronic products.

"The low heat dissipation in Class D amplifiers is taken care of by copper in the PCB itself in the design," says Awasthy. "For high-power applications like 100W or 200W, you will need a heat-sink with Class D amplifiers. However, the size of the heat-sink will be much smaller than that required by Class AB," adds Tan. Class D is widely used for flat-screen TVs above 102 cm (40 inches).

"Incorporating digital signal processing and power stage on a single die fur-

ther allows you to save space in the new Class D power amplifier ICs," says Luigi Pagotto, product marketing manager-audio division, STMicroelectronics.

Thermal protection

Many a times, the temperature in and around a circuit may rise due to inefficiency or malfunctioning of the component. It may damage your expensive electronic product and in worse case cause fire. So thermal protection becomes an important aspect of system design. To protect audio sub-system and the IC itself, Class D amplifiers come with auto-recovery and variable gain control features.

"For some reason if the audio amplifier is exposed to a temperature above 150°C (as rated for thermal shut-down), it will go into a shutdown mode. After the temperature cools down, the part will come up automatically. This is called auto-recovery. This feature has been introduced to avoid thermal runaway of the circuit," explains Awasthy. This is quite useful in avoiding damage to the IC. Similarly, in the case of over-current due to short-circuit in the output stage, the part will go into shutdown mode and recover when the condition becomes normal.

"An alternative solution to shutting down the amplifier is gradually decreasing the gain of the amplifier to a level where the combination of dissipation and thermal resistance from junction to ambient results in a junction temperature just around the threshold level. That means the amplifier will not completely switch off but remain operational at lower output power levels. This is a feature we call thermal fold-back," says Tan.

EMI suppression and noise reduction

Pulse-width modulation in Class D amplifiers causes internal switching of the amplifier circuit, resulting in electromagnetic interference (EMI)/electromagnetic compatibility (EMC) issues. It is necessary to suppress EMI within the set limit. Otherwise, it will

Important considerations for audio power amplifiers

1. Output power requirement
2. Available supply voltage and current
3. Input: analogue or digital
4. Cost
5. Size and form factor as per the product design

start interfering with other components.

"Designers have to design their systems such that they meet the FCC requirement for EMI/EMC," says Kristy Lypen, business director, multimedia business unit, Maxim. An L-C filter consisting of an inductor coil and a capacitor can be used to reduce the EMI. However, it increases the cost of the overall system and also makes it bulky. Texas Instruments has introduced inexpensive ferrite beads to replace the costly and bulky inductor coil.

"EMI can be controlled by modulating the frequency of the clock to reduce the energy on a per-hertz basis. This is called spread-spectrum modulation, where the energy is spread around the fundamental frequency," suggests Lypen.

EMI can be further controlled using active-emission limiting. "Active-emission limiting sets the minimum pulse width in the amplifier. It controls the oscillations called 'ringing' on the edge of PWM signal, thereby providing a huge improvement in EMI performance," adds Lypen.

The PWM switching frequency in the audio power amplifier is 300-400 kHz. There is a higher-order harmonics interference. You have to make sure that the design layout is proper and the grounding is separate for small and large signals. Also, the components should be placed close to the amplifier, keeping the lengths of connecting wires short.

Output stage protection

It is important to ensure that the wattage of amplifiers and speakers match.